

17. Last month, houses in a particular suburb sold for (in ascending order):

\$633 000, \$635 000, \$712 000, \$713 000, \$750 000,
\$795 000, \$810 000, \$820 000, \$820 000, \$1 325 000

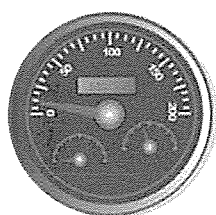


- Find the mean, median and mode of these prices.
- Is the mode typical of last month's sale prices for this suburb?
- How many prices are below the median? the mean?

The mean and median are often close together in a set of scores.

- Are they close together in this case? Suggest a reason.

18. The speed limit near schools is 40 kilometres per hour. A speed camera recorded the actual speed of 25 motorists driving through the school zone. The data is presented in a stem-and-leaf plot.

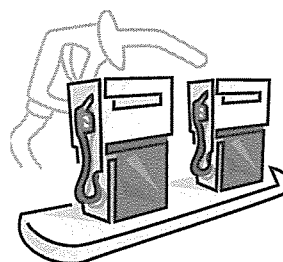


3	5	9	9	8				
4	0	7	5	5	5	8	1	
5	1	0	2	9	7	2	4	4
6	1	3	3	8				
7	0							
8	4							

- What was the highest recorded speed?
- What proportion (%) of motorists broke the speed limit?
- How many motorists broke the speed limit by more than 15 kph?
- Find the mean and median of the distribution of speeds.

19. The stem-and-leaf plot shows the price of petrol (in cents per litre) at a random selection of metropolitan petrol stations, on a Friday.

14	9.3	9.9	9.9		
15	0.1	1.3	2.9	2.9	2.9
16	0.9	0.9	0.5	0.5	
17	0.2	0.5	0.2	0.2	0.2



- How many petrol stations were in the sample?
- What is the highest price for petrol?
- Does this distribution have a mode?
- Find the mean and median price of petrol.

Exercise 6.2 continued

20. The 18 scores below are arranged in ascending order.

5, 6, 6, 7, 8, 10, 11, 11, 12, 14, 15, 15, 16, 17, 19, 20, 20, 22

The median is 13, which is halfway between the 9th and 10th scores. The median also marks the 50th percentile or the second quartile (Q_2).

The median of the lower half of the scores (those below 13), is 8; this value is the lower quartile or the first quartile (Q_1) or the 25th percentile.

The median of the upper half of the scores (those above 13), is 17: this value is the upper quartile or the third quartile (Q_3) or the 75th percentile.

Find the median and the lower and upper quartiles for each of the following sets of data. Remember that the scores must be arranged in ascending order.

- a) 12, 13, 14, 15, 16, 17
- b) 4, 5, 5, 7, 9, 11, 15, 15, 17, 17, 20, 24, 25, 27
- c) 50, 50, 51, 52, 53, 53, 60, 62, 62, 65, 70, 75, 80, 81
- d) 15, 16, 17, 18, 20, 20, 20, 25, 30, 31, 32, 35, 35, 36
- e) 4, 8, 8, 12, 15, 16, 17, 20
- f) 3, 8, 8, 10, 12, 15, 16, 17, 17, 20, 20, 21
- g) 72, 75, 72, 87, 81, 90, 72, 83, 80, 87
- h) 51, 60, 32, 33, 39, 35, 50, 41, 45, 47, 40, 45

21. A group of 16 tourists are travelling on a bus to the Gold Coast. Their ages range from 45 years to 58 years and are presented in the frequency table below.

Age	45	46	47	48	49	50	51	52	53	54	55	56	57	58
f	1	2	0	1	1	2	2	1	1	0	1	2	1	1

- a) Find the mean age of the tourists.
- b) Find the median age of the tourists.
- c) What is the age that twenty five percent of the tourists are younger than?
- d) What is the age that twenty five percent of the tourists are older than?

